

Disproof of conjecture 00000000732:
every unit of $\mathbb{Q}(\zeta_m)$, $m \geq 3$, has norm $+1$, so $\varepsilon(m) = 1$

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1 The conjecture

Statement (English, verbatim). Definition: The unit index $\varepsilon(m)$ of the cyclotomic field $\mathbb{Q}(\zeta_m)$ is the index of its unit group over the norm-one units. Conjecture: As the conductor $m \rightarrow \infty$, the supremum of $\varepsilon(m)$ grows like $\varphi(m)/2^{\omega(m)} \cdot (1 + O(2^{-\omega(m)}))$, with exact equality when m is an odd prime power. (The Chinese text says the same.)

Reading. Let $K = \mathbb{Q}(\zeta_m)$, $E = \mathcal{O}_K^\times$ its unit group and $E_1 = \{u \in E : N_{K/\mathbb{Q}}(u) = 1\}$ the norm-one units. Then $\varepsilon(m) = [E : E_1]$. Here $N_{K/\mathbb{Q}}$ is the absolute norm, φ is Euler's totient, and $\omega(m)$ is the number of distinct prime divisors of m . The conjecture is a conjunction: (A) an asymptotic statement, and (B) the exact equality $\varepsilon(m) = \varphi(m)/2^{\omega(m)}$ for every odd prime power m . For $m = 7$, (B) predicts $\varphi(7)/2^1 = 3$.

2 Definitions in Lean

All objects are built from Mathlib (Lean file `Conjecture732/Basic.lean`, namespace `C732`).

- $K = \mathbb{Q}(\zeta_m)$ is Mathlib's `CyclotomicField m Q` and E is $(\mathcal{O}_K)^\times$, the units of the ring of integers.
- `unitNorm K`: $E \rightarrow \mathbb{Q}^\times$, $u \mapsto N_{K/\mathbb{Q}}(u)$, built from Mathlib's `Algebra.norm Q`. Then `normOneUnits K` is its kernel E_1 .
- `unitIndex m` = $\varepsilon(m)$ = `(normOneUnits (CyclotomicField m Q)).index`. Mathlib's `Subgroup.index` is 0 for an infinite index. That case does not arise for ε , which we show is at most 2.
- `omega m` = `m.primeFactors.card`, and `claimed m` = $\varphi(m)/2^{\omega(m)}$, computed in $\mathbb{Q}$ so that no rounding occurs.

3 The key lemma: norms in a totally complex field are nonnegative

Lemma 1 (`norm_nonneg_of_isTotallyComplex`). *Let K be a number field with no real embeddings. Then $N_{K/\mathbb{Q}}(x) \geq 0$ for every $x \in K$.*

Proof. We have $N_{K/\mathbb{Q}}(x) = \prod_{\sigma} \sigma(x)$, the product over all embeddings $\sigma : K \rightarrow \mathbb{C}$ (Mathlib: `Algebra.norm_eq_prod_embeddings`). Group the embeddings by the infinite place w they define. The embeddings above w are exactly φ_w and $\overline{\varphi_w}$ (Mathlib: `mk_eq_iff`). These two are distinct because φ_w is not real (`IsTotallyComplex.complexEmbedding_not_isReal`). So the product over the fibre of w is $\varphi_w(x)\overline{\varphi_w(x)} = |\varphi_w(x)|^2$, and $N_{K/\mathbb{Q}}(x) = \prod_w |\varphi_w(x)|^2 \geq 0$. $\square$

Theorem 2 (`unitIndex_eq_one`). *For every $m > 2$, $\varepsilon(m) = 1$.*

Proof. For $m > 2$, $\mathbb{Q}(\zeta_m)$ is totally complex (Mathlib: `IsCyclotomicExtension.Rat.isTotallyComplex`). A unit u has $|N(u)| = 1$ (Mathlib: `NumberField.isUnit_iff_norm`), and $N(u) \geq 0$ by the lemma, so $N(u) = 1$. Hence $E_1 = E$ (`normOneUnits_eq_top`), and the index is 1. $\square$

For every m (including $m \leq 2$), $\varepsilon(m) \leq 2$ (`unitIndex_le_two`). The reason is that $u \mapsto N(u)$ maps E into $\{\pm 1\}$, and the index of a kernel equals the size of the image (`Subgroup.index_ker`).

4 Refutation of (B), the exact-equality clause

For an odd prime power $q = p^a$ we have $\omega(q) = 1$ and $\varphi(q) = p^{a-1}(p-1)$, so the claimed value is $p^{a-1}(p-1)/2$ (`claimed_prime_pow`).

- `witness_seven`: $\varepsilon(7) = 1$ and `claimed(7) = 3`.
- `three_le_claimed`: every odd prime power $q = p^a \geq 7$ has claimed value at least 3. (If $a = 1$ then $p-1 \geq 6$. If $a \geq 2$ then $p^{a-1} \geq 3$ and $p-1 \geq 2$.)
- `exact_equality_fails`: $\varepsilon(q) \neq \varphi(q)/2^{\omega(q)}$ for every odd prime power $q \geq 7$, an infinite family. (The equality also fails at $q = 5$, where the claimed value is 2, and holds at $q = 3$, where it is 1. Neither small case is needed or formalized.)

Other readings of “the supremum of $\varepsilon(m)$ ”. A supremum of ε -values over any set T of conductors (for example $m' \leq m$, $m' \geq m$, or $\{m\}$) is at most 2 (`sSup_unitIndex_le_two`). So the exact equality fails at every odd prime power $q \geq 7$ for any function bounded by 2 (`exact_equality_fails_of_le_two`).

Other readings of “unit index”, refuted at $m = 7$ and at every odd prime power $q \geq 7$.

- *Hasse’s unit index* $Q = [E : WE^+]$, where W is the group of roots of unity and E^+ the group of units of the maximal real subfield K^+ . This is Mathlib’s `IsCMField.indexRealUnits`, and Mathlib proves it is 1 or 2 (`indexRealUnits_eq_one_or_two`). We use the CM structure of $\mathbb{Q}(\zeta_m)$ for $m > 2$ (`IsCyclotomicExtension.Rat.isCMField`). Since $1, 2 \neq 3$, `hasse_index_ne_claimed` follows.
- *Relative norm* to K^+ , that is $E'_1 = \{u : N_{K/K^+}(u) = 1\}$ (`relNormOneUnits`). If $[E : E'_1] = n$ were finite and nonzero, then $u^n \in E'_1$ for every u . Take $u = \eta$, a fundamental unit of K^+ (Mathlib’s `Units.fundSystem`). Then $N_{K/K^+}(\eta^n) = \eta^{2n}$ (`Algebra.norm_algebraMap`, $[K : K^+] = 2$), so $\eta^{2n} = 1$. This contradicts the fact that η is not torsion (its log-embedding is a basis vector). The unit rank is positive because $\mathbb{Q}(\zeta_q)$ has $\varphi(q)/2 \geq 3$ complex places (`nrComplexPlaces_eq_totient_div_two`). Hence the index is infinite, which is Mathlib’s value 0 (`relNormOneUnits_index_eq_zero`, `rel_index_ne_claimed`).

5 Refutation of (A), the asymptotic clause

Write $f(m) = \varphi(m)/2^{\omega(m)}$ (in $\mathbb{R}$). We formalize two readings for a function S :

- **BigOClaim S**: there are C and N with $|S(m) - f(m)| \leq C f(m) 2^{-\omega(m)}$ for all $m \geq N$. This is the statement $S = f \cdot (1 + O(2^{-\omega}))$.
- **EquivClaim S**: $S(m)/f(m) \rightarrow 1$.

asymptotic_fails: if $S \leq 2$, both readings fail. Proof: take $m = p_1 \cdots p_k$, a product of k distinct primes $\geq \max(N, 11)$ (**family**, using the infinitude of primes). Then $\omega(m) = k$, $\varphi(m) = \prod (p_i - 1)$, and $f(m) \geq 5^k$. Choose k large enough that $C 2^{-k} < 1/2$. Then $|S(m) - f(m)| \geq f(m) - 2 > f(m)/2 > C f(m) 2^{-\omega(m)}$, and $S(m)/f(m) \leq 2/5$. This applies to $S = \varepsilon$ and to every family of suprema $m \mapsto \sup \varepsilon(T_m)$.

6 Main theorem and axiom audit

C732.conjecture732_false proves, as a single conjunction: (i) $\neg \text{Conjecture732}$, where **Conjecture732** is **BigOClaim** $\varepsilon \wedge \forall p a, p \text{ prime}, p \neq 2, a \geq 1 \Rightarrow \varepsilon(p^a) = \text{claimed}(p^a)$; (ii) $\varepsilon(7) = 1$ and $\text{claimed}(7) = 3$; (iii) for every $T \subseteq \mathbb{N}$, $\sup \varepsilon(T) \neq 3$; (iv) Hasse's index of $\mathbb{Q}(\zeta_7)$ is not 3; (v) the relative-norm index of $\mathbb{Q}(\zeta_7)$ is 0 (infinite); (vi) for every $T : \mathbb{N} \rightarrow \mathcal{P}(\mathbb{N})$, both **BigOClaim** and **EquivClaim** fail for $m \mapsto \sup \varepsilon(T_m)$.

#print axioms C732.conjecture732_false reports only **propext**, **Classical.choice** and **Quot.sound**. The file contains no **sorry** and no **native_decide**. It was built with Lean v4.33.1 and Mathlib v4.33.1.

Remark. Under the literal Definition, $\varepsilon(m) = 1$ for every $m \geq 3$, so the conjectured formula does not match the defined object at any odd prime power other than 3.